Day 4 — TNPSC Group 2A Study Material

Part A – General Studies: Everyday Application: Electricity, Magnetism, Light & Sound

READING MATERIAL

This single syllabus clause bundles four physics areas at once. Real coaching-site evidence shows TNPSC tests a mix of 'why does X happen in daily life' (sky's blue colour, bulb filament shape) and direct formula/definition recall (Ohm's law, lens power, speed of sound) — cover both styles.

Q: What is the resistance of a wire after it's stretched to exactly double its original length (volume unchanged)?

A: It becomes 4 times the original resistance. Since $R = \rho L/A$ and volume ($A \times L$) stays constant, doubling L halves A — so $R_{new} = \rho(2L)/(A/2) = 4\times(\rho L/A) = 4R$. A 1-ohm wire becomes a 4-ohm wire.

Q: Which of these equations are correctly stated? (i) $H=V^2It$ (ii) $V=IR$ (iii) $P=VI$ (iv) $F=mV^2$

A: (ii) $V=IR$ (Ohm's Law) and (iii) $P=VI$ (electrical power) are correct. (i) is wrong (the correct heating-effect formula is $H=I^2Rt$, Joule's law) and (iv) is wrong (force is $F=ma$, not mV^2).

Q: Why is soft iron used for magnetic shielding?

A: Soft iron is highly magnetically permeable — it draws in (concentrates) external magnetic field lines into itself, keeping the shielded region inside relatively field-free.

Q: Why is a light bulb's filament made as a 'coiled coil' rather than a straight wire?

A: A coiled-coil shape packs a long filament into a small space and reduces heat loss through convection currents inside the bulb, so the filament reaches a higher, more efficient glowing temperature.

Q: What rule gives the direction of the magnetic field produced by a current-carrying conductor?

A: Commonly called the Right-Hand (Thumb) Rule. One sourced TNPSC-prep answer calls this 'Fleming's right-hand rule' — be aware that, strictly, Fleming's right-hand rule is normally taught for generators/induced EMF, and Fleming's left-hand rule for motor force; the field-direction-around-a-wire rule is conventionally just the Right-Hand Thumb Rule. If this exact MCQ appears, the coaching-site convention expects 'Fleming right-hand rule' as the labelled answer — know both names.

Q: Why is white NOT considered a 'prime' (primary) colour?

A: White light is a mixture of all the primary colours (commonly Red, Green, Blue in the additive light model) — it's a combination, not a primary colour itself.

Q: A lens has power +5D (dioptres). What is its focal length, and what does the positive sign tell you?

A: Focal length = 1/Power = 1/5 m = 0.2 m = 20 cm. The positive sign indicates it's a convex (converging) lens.

Q: What does an achromatic lens do?

A: It corrects chromatic aberration, producing a clear image unaffected by the colour-fringing that a simple lens would otherwise introduce.

Q: Why is the sky blue?

A: Scattering — specifically Rayleigh scattering, where shorter (blue) wavelengths of sunlight scatter more strongly off air molecules than longer (red) wavelengths, spreading blue light across the sky.

Q: In which medium (among vacuum, air, glass, water) is the speed of light slowest?

A: Glass — light slows down more in denser/more optically refractive media, and of these four, glass has the highest refractive index.

Q: Convert a wavelength of 7000 Å (angstroms) to nanometres.

A: 1 Å = 0.1 nm, so 7000 Å = 700 nm. (This corresponds to red light, near the edge of the visible spectrum.)

Q: The Raman Effect (discovered by C.V. Raman) is the scattering of light by which state of matter?

A: Pure liquids — Raman's original Nobel Prize-winning work studied light scattering in liquids.

Q: Who invented the phonograph, and what did it do?

A: Thomas Alva Edison invented the phonograph, a device that played back recorded sound.

Q: What is the formula relating the speed of sound to frequency and wavelength?

A: $v = n\lambda$ (or $v = f\lambda$) — speed equals frequency (n or f) multiplied by wavelength (λ).

Q: How is the 'speed of sound' defined in terms of distance and time?

A: It's the distance travelled by a sound wave in 1 second (i.e., distance/time with time fixed at 1 second), conventionally denoted 'v'.

MOCK TEST (WITH ANSWERS)

1. A wire of resistance 2 ohms is stretched to exactly double its length (volume unchanged). What is its new resistance?

A. 1 ohm

B. 2 ohms

C. 4 ohms

D. 8 ohms ✓

Explanation: Resistance scales by 4× when length doubles at constant volume ($R=\rho L/A$, and A halves when L doubles at fixed volume): new $R = 4 \times R_{old} = 4 \times 2 = 8$ ohms. [medium, authored]

2. Which of the following is the correctly stated form of Ohm's Law?

A. $P = VI$

B. $V = IR$ ✓

C. $H = V^2It$

D. $F = mV^2$

Explanation: $V = IR$ is Ohm's Law. $P=VI$ is a valid power formula too but isn't Ohm's Law itself; the other two are wrong as written. [easy, web-research]

3. Soft iron is preferred for magnetic shielding because it:

A. Repels magnetic field lines

B. Concentrates/draws in magnetic field lines ✓

C. Has no magnetic properties

D. Generates its own opposing field

Explanation: It draws field lines into itself, shielding the enclosed region. [medium, web-research]

4. A light bulb's filament is made as a 'coiled coil' mainly to:

A. Make it look decorative

B. Reduce heat loss by convection and save space ✓

C. Increase electrical resistance to zero

D. Make it easier to replace

Explanation: Compact shape, reduced convective heat loss, higher operating temperature. [medium, web-research]

5. Which of the following is NOT considered a primary (prime) colour of light?

A. Red

B. Green

C. Blue

D. White ✓

Explanation: White is a mixture of the primary colours, not a primary colour itself. [easy, web-research]

6. A lens has a power of +5 dioptres. Its focal length is:

A. 5 cm

B. 0.5 cm

C. 20 cm ✓

D. 50 cm

Explanation: Focal length = 1/Power = 1/5 m = 20 cm. [medium, web-research]

7. An achromatic lens is designed to:

A. Magnify images extensively

B. Produce a coloured image deliberately

C. Correct chromatic aberration for a clear image ✓

D. Block all visible light

Explanation: It corrects colour-fringing (chromatic aberration). [medium, web-research]

8. The blue colour of the sky is caused by:

A. Refraction

B. Reflection

C. Scattering ✓

D. Interference

Explanation: Rayleigh scattering of shorter wavelengths by the atmosphere. [easy, web-research]

9. Among vacuum, air, glass, and water, the speed of light is slowest in:

A. Vacuum

B. Air

C. Glass ✓

D. Water

Explanation: Glass has the highest refractive index of the four listed, so light travels slowest through it. [medium, web-research]

10. 7000 Å (angstroms) is equal to how many nanometres?

A. 7 nm

B. 70 nm

C. 700 nm ✓

D. 7000 nm

Explanation: 1 Å = 0.1 nm, so 7000 Å = 700 nm. [medium, web-research]

11. The Raman Effect refers to the scattering of light by:

A. Pure solids

B. Pure liquids ✓

C. Pure gases

D. Plasma

Explanation: C.V. Raman's discovery concerned scattering in pure liquids. [medium, web-research]

12. Who invented the phonograph?

A. Heinrich Hertz

B. Thomas Alva Edison ✓

C. Ernest Rutherford

D. Ibn Sahl

Explanation: Thomas Alva Edison. [easy, web-research]

Part B – Aptitude & Mental Ability: Lowest Common Multiple (LCM)

READING MATERIAL

LCM is 1 of 10 named Aptitude sub-skills. Real exam evidence shows two distinct shapes: algebraic LCM-from-a-clue problems, and 'remainder' word problems — both are taught below as separate methods.

Q: Real exam (2025): If the LCM of 'a' and 'a+2' is 1300, find the LCM of 'a' and 'a+1'.

A: Since a and a+2 differ by 2, their HCF is either 1 or 2. If HCF=2: LCM = $a(a+2)/2 = 1300 \rightarrow a(a+2) = 2600$. Solving $a^2+2a-2600=0$ gives a=50 (check: 50×52/2 = 1300 ✓). Now find LCM(50, 51): consecutive integers are always coprime (HCF=1), so LCM = 50×51 = 2550.

Q: Method — 'least number leaving given remainders' (remainders differ from the divisors by a constant): Find the least number which, when divided by 16, 18, and 21, leaves remainders 3, 5, and 8 respectively.

A: Check the gap between each divisor and its remainder: 16−3=13, 18−5=13, 21−8=13 — all the same gap! When that happens, the answer is LCM(16,18,21) – 13. LCM(16,18,21) = 1008. Answer = 1008−13 = 995.

Q: Method — 'least number leaving the SAME remainder in every case': Find the least number which, when divided by 6, 7, 8, 9, and 12, leaves the same remainder 1 in every case.

A: When the remainder is identical across all divisors, the answer is LCM(divisors) + remainder. LCM(6,7,8,9,12) = 504. Answer = 504 + 1 = 505.

Q: Method — 'mixed condition' (a common remainder for some divisors, but exact division for another): Find the least number which, when divided by 5, 6, 7, and 8, leaves remainder 3 in each case, but leaves NO remainder when divided by 9.

A: First find the least number leaving remainder 3 for 5,6,7,8: that's LCM(5,6,7,8)+3 = 840+3 = 843. But we also need it divisible by 9. Keep adding LCM(5,6,7,8)=840 repeatedly to 843 until the result is divisible by 9: 843 (no), 1683 (1683÷9=187, yes!). Answer = 1683.

Q: Method — LCM of numbers given as prime factorizations: Find the LCM of (2×3×5×7) and (3×5×7×11).

A: LCM takes the HIGHEST power of every prime that appears in either number. Primes involved: 2,3,5,7,11 — each appears at most once in either factorization, so LCM = 2×3×5×7×11 (just the union of all prime factors, since none repeat).

MOCK TEST (WITH ANSWERS)

1. If the LCM of 'b' and 'b+2' is 180, and their HCF is 2, find the LCM of 'b' and 'b+1'.

A. 342 ✓

B. 380

C. 361

D. 324

Explanation: LCM = $b(b+2)/HCF = 180 \rightarrow b(b+2) = 360 \rightarrow b^2+2b-360=0 \rightarrow b=18$ (check: 18×20/2=180 ✓). Now LCM(18,19): consecutive integers are coprime, so LCM = 18×19 = 342. [hard, authored]

2. Find the least number which, when divided by 12, 15, and 20, leaves remainders 7, 10, and 15 respectively.

A. 55 ✓

B. 60

C. 65

D. 53

Explanation: Gaps: 12−7=5, 15−10=5, 20−15=5 — all equal, so answer = LCM(12,15,20) – 5 = 60 – 5 = 55. [medium, authored]

3. Find the least number which, when divided by 4, 5, and 6, leaves the same remainder 2 in each case.

A. 58

B. 60

C. 62 ✓

D. 64

Explanation: LCM(4,5,6) = 60. Answer = 60 + 2 = 62. [easy, authored]

4. Find the LCM of (2×2×3×5) and (2×3×3×7).

A. 2×2×3×3×5×7 ✓

B. 2×3×5×7

C. 2×2×3×5×7

D. $4\times9\times5\times7$

Explanation: Take the highest power of each prime: 2^2 (from the first), 3^2 (from the second), 5^1, 7^1. LCM = $2^2\times3^2\times5\times7$, which equals option (A) written out as $2\times2\times3\times3\times5\times7$. [medium, authored]

5. The LCM of two consecutive integers n and n+1 is always:

A. $n + 1$

B. n(n+1) ✓

C. $n - 1$

D. Cannot be determined

Explanation: Consecutive integers are always coprime (HCF=1), so their LCM is simply their product, $n(n+1)$. [easy, authored]

Part C – General English: Poem: "A Tragic Story" — William Makepeace Thackeray

READING MATERIAL

A short comic poem about a sage obsessed with moving his pigtail from behind him to in front, who never succeeds. Real coaching-site evidence shows this poem is tested almost entirely as 'identify the figure of speech in this line' — and the pattern is consistent enough to learn as a rule rather than memorizing 12 separate answers.

Q: What is the plot of "A Tragic Story"?

A: A sage ('in days of yore') is bothered that his pigtail (braid) hangs behind him. He spins around — round and round, all day — trying to bring it to the front. No matter how he turns, the pigtail stays faithfully behind him, since it's physically attached to his back and turns with him. He never succeeds.

Q: What is the poem's central theme/lesson?

A: The futility of attempting an impossible task, told humorously — the sage's persistent, theoretical effort is undercut by his lack of practical understanding (the pigtail can never NOT be behind him while attached to his back, no matter which way he spins).

Q: What is the recurring figure of speech used whenever the poem mentions the pigtail still hanging behind him, despite all his efforts?

A: Irony — there's a gap between his persistent, hopeful effort and the physically inevitable outcome (it was always going to stay behind him). Lines like 'But still it hung behind him' and 'Because it hung behind him' are both cited examples.

Q: What figure of speech describes the poem's repeated phrases like 'round and round' and 'right and left and round about'?

A: Repetition — the same words/phrase repeated for emphasis, here mimicking the sage's repetitive, futile spinning motion.

Q: What figure of speech is used in lines like 'There lived a sage in days of yore', 'All day the puzzled sage did spin', and 'He mused upon this curious case'?

A: Alliteration — repeated initial consonant sounds within the line (sage/spin/days; mused/this — note: alliteration specifically repeats STARTING sounds of nearby words, distinguishing it from the whole-phrase Repetition examples above).

Q: Who wrote "A Tragic Story", and is it meant to be read seriously?

A: William Makepeace Thackeray. Despite the title, it's a humorous/comic poem, not an actual tragedy — the 'tragic' framing is itself part of the joke, given how trivial the sage's problem really is.

MOCK TEST (WITH ANSWERS)

1. What was the sage in "A Tragic Story" trying to do?

A. Grow a longer pigtail

B. Move his pigtail to the front ✓

C. Cut off his pigtail

D. Dye his pigtail

Explanation: He spent all day spinning, trying to bring his pigtail around to the front. [easy, web-research]

2. Was the sage ever successful in moving his pigtail?

A. Yes, after many tries

B. Yes, immediately

C. No, it always stayed behind him ✓

D. The poem doesn't say

Explanation: No matter how he spun, the pigtail — attached to his back — always remained behind him. [easy, web-research]

3. Identify the figure of speech: "But still it hung behind him."

A. Simile

B. Personification

C. Irony ✓

D. Hyperbole

Explanation: Irony — his effort can never change a physically inevitable outcome. [medium, web-research]

4. Identify the figure of speech: the repeated use of "round and round" in the poem.

A. Hyperbole

B. Onomatopoeia

C. Anaphora

D. Repetition ✓

Explanation: Repetition — the same phrase repeated to emphasize his repetitive spinning. [medium, web-research]

5. Identify the figure of speech: "All day the puzzled sage did spin."

A. Hyperbole

B. Alliteration ✓

C. Assonance

D. Personification

Explanation: Alliteration — repeated initial sounds (e.g. 'sage'/'spin'). [medium, web-research]

6. What is the overall dominant figure of speech across "A Tragic Story"?

A. Hyperbole

B. Irony ✓

C. Alliteration

D. Personification

Explanation: Irony — the entire poem's comic premise rests on the gap between effort and an unchangeable outcome. [medium, web-research]

7. "A Tragic Story" was written by:

A. Lewis Carroll

B. Robert Louis Stevenson

C. William Makepeace Thackeray ✓

D. Edgar Albert Guest

Explanation: William Makepeace Thackeray. [easy, web-research]

8. Despite its title, "A Tragic Story" is best described as:

A. A genuine tragedy

B. A humorous poem ✓

C. A horror poem

D. A romantic poem

Explanation: It's comic, not tragic — the title itself is part of the joke given how trivial the sage's 'problem' is. [easy, web-research]